

Mathematics Section A

- Let $a, b \in \mathbb{C}$. Let α, β be the roots of the equation $x^2 + ax + b = 0$. If $\beta - \alpha = \sqrt{11}$ and $\beta^2 - \alpha^2 = 3i\sqrt{11}$, then $(\beta^3 - \alpha^3)^2$ is equal to:
(A) 160 (B) 176 (C) 194 (D) 187
- Let the sum of the first n terms of an A.P. be $3n^2 + 5n$. Then the sum of squares of the first 10 terms of the A.P. is:
(A) 10220 (B) 12860 (C) 15220 (D) 19780
- Let A be a 3×3 matrix such that

$$A^T \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \\ 2 \end{bmatrix}, A^T \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}, A \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 4 \\ 4 \end{bmatrix} \text{ and } A \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$$
 If $\det(A) = 1$, then $\det(\text{adj}(A^2 + A))$ is equal to:
(A) 16 (B) 25 (C) 49 (D) 64
- Consider the system of linear equations in x, y, z :

$$x + 2y + tz = 0$$

$$6x + y + 5tz = 0$$

$$3x + t^2y + f(t)z = 0$$
 where $f: \mathbb{R} \rightarrow \mathbb{R}$ is a differentiable function. If this system has infinitely many solutions for all $t \in \mathbb{R}$, then f
(A) is a constant function
(B) is strictly increasing on $\mathbb{R}$
(C) is strictly decreasing on $\mathbb{R}$
(D) has two critical points
- $\sum_{n=1}^{10} \left(\frac{528}{n(n+1)(n+2)} \right)$ is equal to:
(A) 65 (B) 130 (C) 220 (D) 440
- Let $\tan A, \tan B$, where $A, B \in \left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$, be the roots of the quadratic equation $x^2 - 2x - 5 = 0$. Then $20 \sin^2\left(\frac{A+B}{2}\right)$ is equal to:
(A) $10 + \sqrt{10}$ (B) $10 - 2\sqrt{10}$
(C) $10 - 3\sqrt{10}$ (D) $10 - \sqrt{10}$
- A letter is known to have arrived by post either from KANPUR or from ANANTPUR. On the envelope just two consecutive letters AN are visible. The probability, that the letter came from ANANTPUR, is:
(A) $\frac{7}{10}$ (B) $\frac{10}{17}$
(C) $\frac{12}{19}$ (D) $\frac{7}{19}$
- The mean deviation about the mean for the data

x_i	5	7	9	10	12	15
f_i	8	6	2	2	2	6

is equal to:

- (A) $\frac{40}{13}$ (B) $\frac{42}{13}$
(C) $\frac{44}{13}$ (D) $\frac{46}{13}$
- Let a focus of the ellipse $E: \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ be $S(4, 0)$ and eccentricity be $\frac{4}{5}$. If the point $P(3, \alpha)$ lies on E and O is origin, then the area of $\triangle POS$ is equal to:
(A) $12/5$ (B) $14/5$ (C) $24/5$ (D) $48/5$
- Let P be a moving point on the circle $x^2 + y^2 - 6x + 8y + 21 = 0$. Then, the maximum distance of P from vertex of the parabola $x^2 + 6x + y + 13 = 0$ is equal to
(A) 8 (B) 10 (C) 12 (D) 9
- In an equilateral triangle PQR , let the vertex P be $(3, 5)$ and the side QR be along the line $x + y = 4$. If orthocentre of the triangle PQR is (α, β) , then $9(\alpha + \beta)$ is equal to:
(A) 16 (B) 27 (C) 36 (D) 48
- The sum of all the integral values of p such that the equation $3 \sin^2 x + 12 \cos x - 3 = p, x \in \mathbb{R}$, has at least one solution, is:
(A) -54 (B) -60 (C) -75 (D) -84
- The square of the distance of the point $P(5, 6, 7)$ from line $\frac{x-2}{2} = \frac{y-5}{3} = \frac{z-2}{4}$ is equal to:
(A) 3 (B) 5 (C) 6 (D) 8
- Let $\vec{a} = \sqrt{7}\hat{i} + \hat{j} - \hat{k}$ and $\vec{b} = \hat{j} + 2\hat{k}$. If $\vec{r}$ is a vector such that $\vec{r} \times \vec{a} + \vec{a} \times \vec{b} = \vec{0}$ and $\vec{r} \cdot \vec{a} = 0$, then $|3\vec{r}|^2$ is equal to:
(A) 44 (B) 54 (C) 86 (D) 132
- The square of the distance of the point of intersection of the lines $\vec{r} = (\hat{i} + \hat{j} - \hat{k}) + \lambda(a\hat{i} - \hat{j})$, $a \neq 0$ and $\vec{r} = (4\hat{i} - \hat{k}) + \mu(2\hat{i} + a\hat{k})$ from the origin is:
(A) 5 (B) 10 (C) 17 (D) 26
- The area of the region $R = \{(x, y) : xy \leq 27, 1 \leq y \leq x^2\}$ is equal to:
(A) $78 \log_e 3 - \frac{52}{3}$ (B) $54 \log_e 3 - \frac{52}{3}$
(C) $54 \log_e 3 - \frac{26}{3}$ (D) $54 \log_e 3 + \frac{26}{3}$
- The product of all possible values of α , for which $\lim_{x \rightarrow 0} \left(\frac{1 - \cos(\alpha x) \cos((\alpha+1)x) \cos((\alpha+2)x)}{\sin^2((\alpha+1)x)} \right) = 2$, is:

- (A) -2 (B) 1
(C) -1 (D) $\frac{5}{4}$

18. The value of the integral $\int_0^\infty \frac{\log_e(x)}{x^2+4} dx$ is:

- (A) $\frac{\pi \log_e(2)}{2}$ (B) $\frac{\pi \log_e(2)}{4}$
(C) $1 + \pi \log_e(2)$ (D) $2 + \pi \log_e(2)$

19. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a differentiable function such that $f\left(\frac{x+y}{3}\right) = \frac{f(x)+f(y)}{3}$ for all $x, y \in \mathbb{R}$, and $f'(0) = 3$. Then the minimum value of the function $g(x) = 3 + e^x f(x)$, is:

- (A) $3\left(\frac{e+1}{e}\right)$ (B) $3\left(\frac{e-1}{e}\right)$
(C) $\frac{3-e}{e}$ (D) $3e$

20. The value of the integral $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \left(\frac{4 - \operatorname{cosec}^2 x}{\cos^4 x}\right) dx$ is:

- (A) $\frac{11}{\sqrt{3}}$ (B) $\frac{16}{\sqrt{3}}$
(C) $\frac{32}{3\sqrt{3}}$ (D) $\frac{64}{3\sqrt{3}}$

Mathematics Section B

21. Let $A = \{1, 2, 3, 4, 5, 6\}$. The number of one-one functions $f: A \rightarrow A$ such that $f(1) \geq 3, f(3) \leq 4$ and $f(2) + f(3) = 5$, is ____.

22. Two players A and B play a series of games of badminton. The player, who wins 5 games first, wins the series. Assuming that no game ends in a draw, the number of ways, in which player A wins the series is ____.

23. If the sum of the coefficients of x^7 and x^{14} in the expansion of $\left(\frac{1}{x^2} - x^4\right)^n, x \neq 0$, is zero, then the value of n is ____.

24. If $\frac{\pi}{4} + \sum_{p=1}^{11} \tan^{-1}\left(\frac{2^{p-1}}{1+2^{2p-1}}\right) = \alpha$, then $\tan \alpha$ is equal ____.

25. Let $y = y(x)$ be the solution of the differential equation $x \sin\left(\frac{y}{x}\right) dy = (y \sin\left(\frac{y}{x}\right) - x) dx, y(1) = \frac{\pi}{2}$ and let $\alpha = \cos\left(\frac{y(e^{12})}{e^{12}}\right)$. Then the number of integral value of p , which the equation $x^2 + y^2 - 2px + 2py + \alpha + 2 = 0$ represents a circle of radius $r \leq 6$, is ____.

Physics Section A

26. In a Vernier calipers, when both jaws touch each other, zero of the Vernier scale is shifted to the right of zero of the main scale and 7th Vernier division coincides with a main scale reading. If the value of 1 main scale division is 1 mm and there are 10 Vernier scale divisions, then the Vernier caliper has

- (A) 0.07 cm negative zero error (B) 0.7 cm negative zero error
(C) 0.07 cm positive zero error (D) 0.7 cm positive zero error

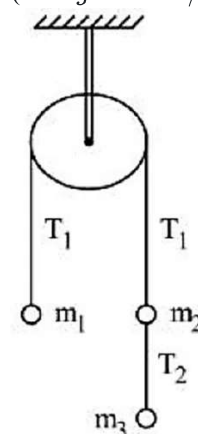
27. L, C and R represents physical quantities inductance, capacitance and resistance respectively. The dimensional formula $ML^2 T^{-4} A^{-2}$ corresponds to ____.

- (A) $\frac{R}{\sqrt{LC}}$ (B) $\frac{R}{LC}$
(C) $\frac{C}{\sqrt{LR}}$ (D) $\frac{1}{R} \sqrt{\frac{L}{C}}$

28. When one moves from a point 16 km below the earth's surface to a point 16 km above the earth's surface. The change in g is approximately $\alpha\%$. The value of α is ____.
(Take radius of the earth = 6400 km.)

- (A) 0.12 (B) 0.25 (C) 0.50 (D) 0.75

29. Three masses $m_1 = 4$ kg, $m_2 = 4$ kg and $m_3 = 6$ kg suspended from a fixed smooth frictionless pulley shown in the figure below. The value of T_1/T_2 is ____ (take $g = 10$ m/s²)



- (A) 5/3 (B) 2/3
(C) 3/5 (D) 2/5